

# Assignment #1: Containers with Docker

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GitHub: <https://github.com/arham-hi/ITCS6190-A1-Arham>

## **Commands Used**

From the project directory, with Docker Desktop running:

```
docker compose up --build
```

```
Get-Content .\out\summary.json
```

```
docker compose down
```

## **Summary from Standard Output (Contents of out/summary.json)**

The app printed === Summary === followed by this JSON. Whitespace has been condensed to fit this report.

```
{
  "total_trips": 12,
  "avg_fare_by_city": [
    {"city": "Atlanta", "avg_fare": 21.75},
    {"city": "Charlotte", "avg_fare": 16.25},
    {"city": "Dubai", "avg_fare": 33.75},
    {"city": "Lisbon", "avg_fare": 14.0},
    {"city": "London", "avg_fare": 38.0},
    {"city": "New York", "avg_fare": 19.0},
    {"city": "San Francisco", "avg_fare": 20.25}
  ],
  "top_by_minutes": [
    {"city": "Dubai", "minutes": 35, "fare": 42.0},
    {"city": "London", "minutes": 35, "fare": 38.0},
    {"city": "Atlanta", "minutes": 28, "fare": 26.0},
    {"city": "San Francisco", "minutes": 28, "fare": 29.3},
    {"city": "New York", "minutes": 26, "fare": 27.1},
    {"city": "Dubai", "minutes": 22, "fare": 25.5},
    {"city": "Charlotte", "minutes": 21, "fare": 20.0},
    {"city": "Atlanta", "minutes": 18, "fare": 17.5},
    {"city": "Lisbon", "minutes": 16, "fare": 14.0},
    {"city": "Charlotte", "minutes": 12, "fare": 12.5}
  ]
}
```

## **Reflection**

I learned how Docker Compose connects a Python application and a PostgreSQL database using the database service name, db, as the hostname. I learned how environment variables supply connection settings and how a bind mount makes the container's output available on my computer. An initial connection error helped me understand that Docker Desktop's engine must be running before Compose can build and start containers. I also learned how database health checks and connection retries help coordinate service startup. I would improve the project by testing a larger dataset with tied trip durations and adding automated checks for the query results.